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Section A: Executive Summary & Introduction

We propose to design and build an automated powder packaging line that uses OPP/CPP plastic film and an ESP32-based control system.

The system will form bags from film, fill each with a precise powder dose, heat-seal the bag, and cut it, achieving at least 15 bags per minute throughput and the approximate weight will be between the following two Values 100 gram - 500 gram. This project builds on modern packaging technology For Example, Vertical Form Filling Seal (VFFS) machines create bags from film rolls and adapts it to our scale and specifications.

The Machine will include a film unwind, forming tube, filling auger, load cell weigh scale, heating sealing jaws, and an ESP32 microcontroller orchestrating sensors and motors.

All key design choices (material type, components, control) reflect discussions to date and align with the stated objectives.

1.1 Key Objectives

- **Automated Bagging:** Build a VFFS machine for powder, using OPP/CPP film as the packaging material (these polypropylene films offer high strength and moisture resistance).
- **Throughput Target:** Achieve a production speed of ≥ 15 bags/min (900 bags/hr), matching Medime industrial baggerr.
- **High-Precision Filling:** Develop an auger filler system capable of dispensing powder with high accuracy.
- **Weight Accuracy:** Pack a precise target weight per bag (e.g. 0.5 kg of salt) with minimal variance to ensure a precision weighing system to control fill quantity.
- **Control System:** mplement an ESP32 microcontroller to automate sequencing (film feed, weighing, sealing) and user interface (start/stop controls).
- **Cost Constraint:** Complete the design within a budget of \\$1000 USD, covering all materials, electronics, and assembly.

1.2 Production Line Features

- **Film Handling:** A roll of OPP/CPP film is unwound and formed into a tube via a forming collar. OPP (uniaxially oriented PP) film is selected for its clarity and strength; CPP (cast PP) may be used as a sealable inner layer due to its excellent moisture barrier.
- **Filling System:** A calibrated auger dispenses powder into each forming bag, ensuring each pouch meets the target weight.
- **Sealing Mechanism:** Heated sealing bars (powered by a controlled heater) will vertically seal the film edges and horizontally seal/cut each bag. The high-melting-point properties of polypropylene allow reliable heat-sealing.
- **Actuators & Sensors:** Stepper motors (with drivers) advance the film and operate to rotate the blades and auger, while sensors (limit switches or photoeyes) detect film position. Relays will switch the heater and motor drivers under ESP32 control. This reflects the block diagram approach of similar systems.
- **Control Electronics:** The ESP32 board reads inputs (start button, sensors) and executes the bagging sequence in software. It controls outputs (motor drivers, heater relay) to coordinate tasks automatically. An display(HMI) or connectivity module provide status.
- **Frame and Hygiene:** All product-contact parts will use safe materials (e.g. stainless steel hoppers,..), and the frame will support easy cleaning. The enclosed design minimizes contamination.

1.3 Financial Overview

- **Budget:** The total budget is set at \ \$1000 USD. This must cover all components: film stock, mechanical parts (rollers, frame), motors, sensors (load cell, photo-sensors), ESP32 board, power supply, wiring, and labor.
- **Cost Allocation:** Electronics (ESP32, drivers, sensors) cost a few hundred dollars; mechanical elements (motors, frame, sealing bars) use the remainder.

1.4 Summary

In summary, this proposal outlines an integrated packaging line for powder that meets the previously discussed requirements.

By combining established VFFS technology with modern IoT control, the system will reliably produce at least 15 sealed bags per minute, each with a precise salt dosage. The design is consistent and compatible with our earlier design discussions (film type OPP/CPP, use of ESP32, component list, etc.) and is structured to deliver the stated outputs within the specified budget.

Section B: Background and Problem Analysis

2.1 Historical Background

Automated packaging machinery has evolved significantly since the mid-20th century. The first vertical form-fill-seal (VFFS) machine was introduced in 1959, using a draw-down belt mechanism to form, fill, and seal bags, a principle still used today. Over the decades, improvements in film materials accompanied the machinery advances. Polypropylene films (like OPP/CPP) became popular since the 1970s for food packaging due to their superior clarity and moisture barrier. Biaxially-oriented PP (BOPP) and uniaxially-oriented PP (OPP) films allowed transparent, high-strength bags. Modern machines now reach extremely high speeds (hundreds of bags per minute), though our design targets a modest 15 bags/min to suit project scope. This history of incremental innovation from basic film extruders to today's servo driven VFFS line provides the foundation for our design.

2.2 Problem Definition

The local powder-packaging sector in Yemen faces a persistent capability gap: small and medium producers largely depend on imported packaging machinery that is costly to acquire, difficult to maintain, and frequently incompatible with local operating conditions (dusty environments, variable grid power, and specific powder characteristics). As a result, packing operations are often manual or semi-automated, which produces low and inconsistent throughput, poor weight accuracy, frequent production stoppages for maintenance or spare parts, and higher unit packaging costs than achievable with localized automation.

These constraints directly lower factory productivity, increase product loss (under-/over-filling and film waste), and reduce competitiveness of domestic powder producers in local and regional markets.

Key problems include:

- **Low Throughput:** Current methods often produce far less than the desired target (15 bags/minute), thus limiting production capacity and revenue.
- **Weight Inaccuracy:** Human filling introduces significant variance in bag weight, Lack of precise dosing can cause underfill or overfill.
- **Environmental and product-handling issues:** Like powdered salt tends to bridge, rat-hole, and create airborne dust, without purpose-designed hoppers, anti-bridging measures, and dust extraction, feed reliability and worker safety suffer.
- **Poor seals on OPP/CPP.**
- **Energy vulnerability:** Many imported solutions assume stable grid power and higher energy budgets, local facilities often need lower-power designs.

2.3 Overall and Specific Objectives

- **Overall Objective:**

Develop a fully automated form-fill-seal packaging machine for powder using OPP/CPP film, capable of high speed, accurate bagging.

- **Specific Objectives:**

- Achieve a steady output of **≥ 15 bags/min** with clean seals.
- Use an **ESP32 microcontroller** to automate the sequence and allow easy adjustments.
- High-Precision Filling, Develop an auger filler system capable of dispensing powder with high accuracy to ensure a **precision weighing system** (load cell) to control fill quantity.
- Employ durable, food safe materials (stainless steel,...) in all contact parts.
- Modular Design, Ensure parts can be easily replaced or repaired using locally available tools.

2.4 Targeted Activities and Outputs

Planned activities:

- **Design & Engineering:** Create mechanical and electrical schematics. Select components (film feed, motors, sensors, etc.) based on our discussions.
- **Procurement:** Acquire all required parts (OPP/CPP film rolls, ESP32 board, motor drivers, sensors, heaters, framing materials).
- **Assembly:** Build the machine frame, install film unwind system, forming tube, conveyor, hopper, and sealing jaws. Wire the ESP32, connect drivers and relays.
- **Software Development:** Program the ESP32 to control the process (start/stop sequence, motor actuation, heater timing, weight feedback).
- **Testing & Calibration:** Run trials with salt to calibrate dosing and sealing temperature. Adjust timings and load-cell calibration for consistent bag weights.
- **Documentation:** Prepare user manual, maintenance guide, and the final report/presentation.

Expected outputs:

- A **working prototype** packaging line .
- Demonstrated **production capability** of ≥ 15 bags/min and correct weight per bag.
- Project report including design drawings, code listings, test data (e.g. consistency of bag weights).
- A final presentation or PDF summarizing the system (mechanical drawings, control diagrams, etc.).

2.5 Beneficiaries

- **Powder Producers/Manufacturers:** Gain a reliable, automated packing solution that boosts capacity and reduces labor costs.
- **End Consumers/Retailers:** Receive consistently filled, well sealed salt bags, improving product quality and shelf life.
- **Engineering Team/Students:** Develop our skills in mechatronics, IoT control, and packaging technology programming, and calibrating the VFFS Auger Filler provides essential, multidisciplinary training through this project.
- **Local Industry and Economy:** Encourages local innovation by providing a low-cost packaging machine, and can be adapted for other granular products.

2.6 Impact and Impact Pathways

The project's implementation will have several positive impacts:

- **Productivity:** Meeting no less 15 bags/min significantly effectively multiplying production rates.
- **Quality Control:** Automated weighing ensures every bag meets target weight, reducing waste from under/over filling.
- **Cost Efficiency:** Automated processes will lower long term labor and material waste costs. Precise cutting and bagging minimizes film waste.
- **Knowledge and Technology Transfer:** Demonstrates how IoT controllers (ESP32) can be applied in manufacturing, which can inspire further automation projects.
- **Scalability:** Although designed for specific powder product types, the line's principles can extend to other powders/granules, multiplying its benefit in the food industry.

2.7 Relevance

This project is highly relevant to industry needs and educational goals. It aligns with modern **Industry 4.0** trends by integrating sensors and IoT control (ESP32) for real-time automation help us for develop our skills in mechatronics, and packaging technology programming, and calibrating the VFFS Auger Filler provides essential, multidisciplinary.

Using OPP/CPP packaging is standard in the food sector, so our system meets actual market requirements.

At the same time, the project fits within an **academic/engineering curriculum** by covering mechanical design, electronics, and control software.

Despite the constrained budget, the project remains a feasible small-scale venture, demonstrating how even limited resources can realize advanced manufacturing solutions.

Section C: Methodology and Timeline

3.1 Methodology

We will employ a classic **Vertical Form-Fill-Seal (VFFS)** approach.

First, the film is unwound from a roll and guided through rollers to a forming collar. A tube is created as the film wraps around a vertical form tube, and the overlapping edges are sealed vertically by a heated bar. The ESP32-based controller then activates the lower sealing jaw to form the bottom of the bag.

Next, powder is dispensed: an actuator (e.g. a stepper/servo driven auger) feeds salt into the newly-formed pouch. A load cell under the fill chute measures the weight in real time, and the ESP32 uses this feedback to stop filling precisely at the target weight. Once filled, a heated horizontal sealer closes the top and simultaneously cuts the bag off. Sensors (photodetector or limit switches) and timed control ensure proper film indexing. Throughout, the ESP32 reads inputs and controls outputs (motors, relays) as outlined in a typical automation block diagram.

The sequence of operations is: *Start* → *Feed Film* → *Seal Bottom* → *Feed Salt* → *Seal & Cut* → *Dispense Bag*, looping continuously. Each mechanical and electrical component is selected and tested to meet the specifications (e.g. motor torque for film feed, heater wattage for timely seal).

Key methodological steps:

- **Mechanical Design:** Draw and fabricate the machine frame, film unwind shaft, forming tube, feed conveyor, and seal jaws. Ensure adjustability for bag size.
- **Weighing System:** Integrate a calibrated load-cell under the filling area. Use an amplifier and ADC to interface its signal to the ESP32.
- **Actuation:** Use stepper or DC motors with drivers to advance film and drive any conveyors. Use relays or MOSFETs to power the sealing heaters.
- **Control Software:** Write the ESP32 firmware (python) to manage the state machine, read sensors, and actuate devices. Include safety checks and an emergency stop.
- **Testing & Iteration:** Perform iterative tests: verify bag integrity (seal strength), weight accuracy (adjust auger time or motor steps), and consistent cycle timing. Optimize code parameters (e.g. heater dwell time, motor speed).

3.2 Preliminary Timeline

1. **Weeks 1–2 (Measurements & Calculation):** Finalize requirements, material specs (OPP/PPP), bag sizes (500–2000 g), auger sizing calculations, load-cell spec and power budget. Deliverable: Requirements & Calculation Report.
2. **Weeks 3–5 (Engineering Design & Simulation):** Mechanical CAD (frame, hopper, auger, forming tube, seal jaws), electrical schematics (ESP32, drivers, SSRs, load cell interface), dynamic simulations and sealing/heat budget checks. Deliverable: Full design package (CAD + electrical drawings + BOM).
3. **Weeks 6–8 (Assembly & Mechanical Installation):** Found of frame and auger, mounting forming tube, installation of sealing jaws, film unwind and dancer, and mechanical integration. Deliverable: Assembled mechanical subsystem.
4. **Weeks 9–10 (Firmware Development & Calibration):** ESP32 firmware (state machine, closed-loop fill control), motor tuning, load-cell calibration, heater PID setup, HMI parameters. Deliverable: Firmware & calibration tables.
5. **Weeks 11–12 (Testing, Optimization & Handover):** Pilot runs with salt, throughput and accuracy verification (≥ 15 bags/min, $\pm 1.5\%$), seal quality checks, corrective tuning, documentation, and complete documentation and presentation materials. Deliverable: Final test report, O&M manual, acceptance sign-off.

Section D: Conclusion and References

4.1 Conclusion

In conclusion, this proposal presents the design and implementation plan for a powder packaging machine that combines proven VFFS mechanics with a precision auger filler and an ESP32-based control system. The solution is optimized for powder packaged in OPP/CPP film (preset weights 100 g – 500g), and balances performance, robustness and cost-effectiveness within the stipulated US\$1,000 budget. The machine's modular architecture (film unwind → forming tube → auger dosing → load-cell feedback → thermal sealing & rotary cutting → discharge) enables straightforward fabrication, commissioning and future upgrades. The project plan, methodology and 12-week timeline map all critical steps from calculations and CAD design through assembly, firmware calibration and pilot testing, ensuring pragmatic execution and rapid validation.

The machine's ESP32 microcontroller automates the entire process — film feed, bag forming, filling, sealing, and cutting — while maintaining cost-effectiveness and reliability in unstable power conditions.

Key expected outcomes include:

- Achieving ≥ 15 bags per minute throughput.
- Filling accuracy target $\pm 1.5\%$ across weight presets.
- Energy-efficient operation (≤ 3 kW).
- Full local assembly, calibration and maintainability.
- Deliverables: working prototype, CAD & electrical drawings, firmware source, test reports.

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